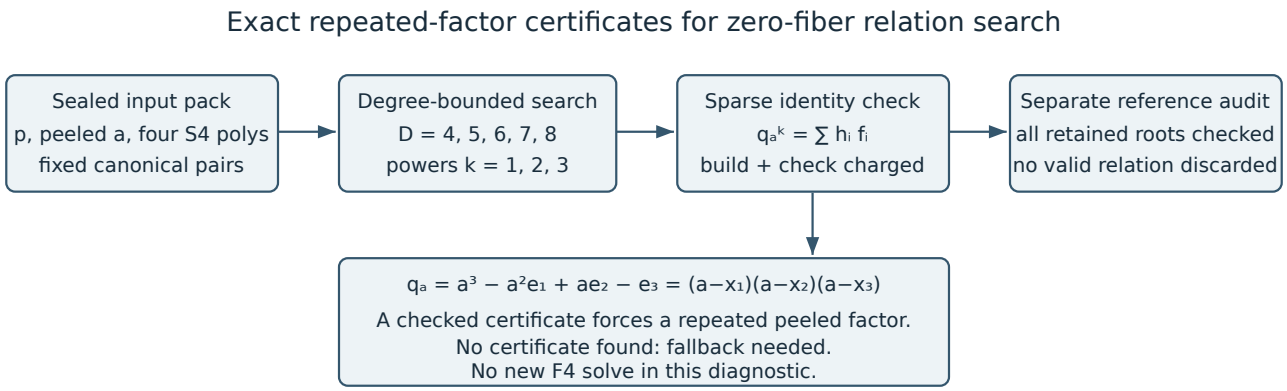
Collision certificates: construction and proof

Implementation checkpoint, 25 October 2026. Actual case comparison pending.

The existing mixed-component zero-fiber collector solves four restricted S4 polynomials in three elementary symmetric coordinates. This follow-up tests whether a small exact certificate can prove that every solution repeats the ordinary factor already peeled from the relation. Such a solution cannot pass the existing distinct-factor decoder.



Matched phi and doubling; 42 case-arms; first plus repeat; complete pipeline price remains open.

The sufficient rejection rule

Let $q_a = a^3 - a^2 e_1 + a e_2 - e_3$ and let the input equations be f_0, f_1, f_2, f_3. The producer searches for polynomial multipliers with $q_a^k = \sum_{i=0}^{3} h_i f_i$. At a common zero, $q_a^k = 0$; in a field this implies $q_a = 0$. Substitution of the three elementary symmetric expressions gives $(a - x_1)(a - x_2)(a - x_3) = 0$, so some factor abscissa equals the peeled a. This is a sufficient condition. A degree-capped failure leaves the system requiring fallback.

The dense Macaulay reduction searches total degrees 4 through 8 and powers 1,2,3 in fixed order. A separate sparse expansion checks the coefficient identity over the actual prime field. Both construction and checking are charged, including unsuccessful degrees. Original roots and relation hits are excluded from the producer's input pack and supplied to the later audit.

Checked implementation controls

Gate v3: five Rust tests passed, zero failed, production build and Lean exit 0. Controls include a linear certificate, a nonradical system requiring a square, a distinct-root system where rejection must fail, a tampered coefficient, duplicate/noncanonical terms, the Vieta identity over the toy field, and the Phase 1 schema regression. These are implementation checks, not a measured benefit on the 42 case-arms.

Native executable SHA-256:
4c49d77faa9b5ef3fd54459bd0ac0cf69ee077eef43ec84361753f7dbc3555f5

Scope of the proof and comparison

Lean 4.34.1, compiler commit `5045d0056413266e57c625dcd7c365b10e377c52`, checks `RepeatedFactor.cubic_identity`, `power_zero` and `rejection_of_certificate`. The rejection theorem exposes an explicit no-zero-divisor hypothesis and an evaluated certificate identity. The cubic and rejection lemmas use `propext`, `Classical.choice`, `Quot.sound`; `power_zero` has no axioms. No accepted declaration uses `sorryAx`.

The native parser, actual finite-field representation, polynomial evaluation bridge, S4-to-group bridge and decoder implementation are not formalized by these lemmas. The independent expansion and retained physical references check those finite executions within their stated scope. This is a conditional algebraic rejection proof, with executable integration obligations still open.

The universal cubic identity has standard certificate `IDC1h10810f17e0f6c8f6`, with 64 derived and 64 extra evaluation points and a hash pointer to the checked Lean source. A false identity obtained by adding one is refused. That standard certificate's fixed Mersenne modulus does not replace each instance's sparse expansion over its own small prime.

Registered experiment

All seven original primes, all three presentations, both phi and doubling: 42 case-arms, each with a first execution and one deterministic repeat (84 producer calls), 42 independent first-run audits and 42 parity checks. The first two p37 seed-0 case-arms also serve as the actual implementation gate and are retained in the full comparison. No extra pilot is repeated. All canonical pairs are included, with their original polynomial inputs.

Each case has one worker, 8 GiB, a 1,200-second outer watchdog and a 1,180-second producer deadline. Separate reservations let the already running 85-configuration target-recovery panel continue. Actual launch receipts are written after the reservation wait. Existing successful 189+21 and Phase 1 F4/point-census studies are reused, never rerun.

Failures retained and price boundary

Gate v1: four Rust tests/build pass; the auxiliary Lean zero-exponent branch fails. Gate v2: four Rust tests/build and corrected Lean pass. Preparation v1 rejects a valid completed Phase 1 record because the new parser expected a pipeline status tag. The parser is corrected and covered by gate v3′s fifth test. None of these attempts runs a new retained-system F4 solve.

The production result remains pending at this checkpoint. The comparison will report certified system counts, successful/failed degree work and verification cost. Any replacement calculation combines actual certificate multiplications with retained F4 row counts on uncertified systems; it is a partial stage model. Other F4 work, polynomial construction, memory, hashing and I/O remain unpriced. No new fallback execution, integrated target/rank result, admitted timing, complete-cost gain or rho claim follows. Canonical IC/rho charts and the benchmark index stay unchanged until their existing gates are met.

Sources: `collision_certificate/PROTOCOL.md`, `EXECUTION.md`, `proof/PROOF_MAP.md`; raw native gates v1/v2/v3, preparation v1/v2 and `collision-certificate-identity-v1`. Standard prior symmetrization work: Faugère et al., EUROCRYPT 2014 (publisher abstract read; full text unavailable).